

Bayero University Kano

Faculty of Computing and Information Technology

CSC 4311: Artificial Intelligence

Module 4: Knowledge Representation and Reasoning

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4.1 Introduction to Knowledge Representation

Definition

Knowledge Representation (KR) is the process of encoding information about the world into a structured form that an AI system can store, process, and reason about. It bridges the gap between **raw data** and **intelligent decision-making**, turning facts into forms the AI can manipulate.

Definition

Knowledge Representation (KR) is the process of transforming real-world information into a formal, machine-readable structure that an Artificial Intelligence system can store, interpret, and use to make decisions. It is essentially the **language of thought** for AI — a way of describing the world so that machines can “understand” it in a logical, consistent, and actionable manner.

Historically, KR emerged as a core branch of AI research in the 1970s and 1980s, focusing on how to represent facts, concepts, and relationships in such a way that a computer could **reason** about them, much like a human would. Without KR, an AI system is like a person with access to a library full of books in a language they cannot read — the knowledge is there, but it cannot be used effectively.

In AI, KR is more than just storing data; it involves encoding **meaning** and **context**. For example, the raw fact “*Paris*” becomes meaningful when represented as: “*Paris is the capital of France*”, which can be linked to other facts, such as “*France is in Europe*”.

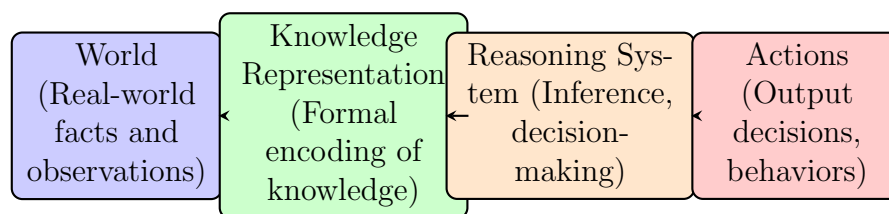
This interconnection allows reasoning systems to derive new information, e.g., “*Paris is in Europe*”.

An everyday analogy: Imagine teaching a robot about your kitchen. Simply showing it objects (data) is not enough. You must also represent:

- **What** the objects are (cup, plate, stove).
- **Where** they are located.
- **How** they are used.
- **Rules** about them (“Do not put metal in the microwave”).

Only with such structured and contextualized knowledge can the robot act intelligently.

In summary, KR acts as the **bridge** between perception (what the AI observes) and reasoning (what the AI concludes or decides to do). It provides the formal framework that makes intelligent behavior possible.



Why Knowledge Representation is Important

1. **Understanding the World** – Gives AI the ability to model entities, events, and relationships.
2. **Reasoning** – Allows inference of new knowledge from existing facts.
3. **Communication** – Medium for humans and machines to share and understand information.
4. **Problem-Solving** – Helps AI agents plan and decide actions based on knowledge.

Types of Knowledge Representation

Knowledge can be represented in different ways depending on the nature of the domain, the reasoning requirements, and computational constraints. The four most common types are:

1. Logical Representation

Logical representation uses formal logic to express knowledge as facts and rules. It is precise, unambiguous, and allows formal reasoning.

- **Propositional Logic** – Deals with statements that are either true or false.

Examples of Propositional Logic

Propositional logic can be applied to a wide variety of real-world situations. Below are some illustrative examples:

Example 1: Weather Prediction Let:

P : “It is raining”, Q : “The ground is wet”

Logical statement:

$P \rightarrow Q$ (If it is raining, then the ground is wet.)

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Example 2: Classroom Attendance Let:

P : “Student attends class”, Q : “Student takes the test”

Statement:

$P \wedge Q$ (The student attends class AND takes the test.)

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Example 3: Lab Door Access Let:

P : “You have a key”, Q : “The lab door is open”

Statement:

$P \vee Q$ (You have a key OR the lab door is open.)

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Example 4: Digital Circuit Logic (OR Gate) Let:

P : Input 1 is HIGH, Q : Input 2 is HIGH

An OR gate outputs HIGH if at least one input is HIGH:

$$P \vee Q$$

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

- **First-Order Logic (FOL)** – Extends propositional logic by including objects, predicates, and quantifiers.

Examples of First-Order Logic (FOL)

First-Order Logic allows us to express statements involving objects, their properties, and relations between them. Below are symbols used in (FOL):

Symbols Used in First-Order Logic

Symbol	Name	Meaning / Usage
$\wedge$	Conjunction	Logical AND (both must be true)
$\vee$	Disjunction	Logical OR (at least one is true)
$\neg$	Negation	Logical NOT (opposite truth value)
$\rightarrow$	Implication	If ... then ...
$\leftrightarrow$	Biconditional	If and only if (IFF)
$\forall$	Universal Quantifier	“For all” / “For every”
$\exists$	Existential Quantifier	“There exists” / “For some”
$=$	Equality	Two terms are identical
$\neq$	Inequality	Two terms are different
$P(x)$	Predicate	Property or relation involving x
$R(x, y)$	Predicate	Relation between x and y
$f(x)$	Function	Maps object x to another object
c	Constant	Represents a specific object
x, y, z	Variables	Represent arbitrary objects

Example 1: All Humans are Mortal Natural language:

“All humans are mortal.”

$$\forall x (Human(x) \rightarrow Mortal(x))$$

Meaning: For every object x , if x is a human, then x is mortal.

Example 2: Socrates is a Human Natural language:

“Socrates is a human.”

$$Human(Socrates)$$

Meaning: The individual “Socrates” belongs to the category of humans.

Example 3: There Exists a Student Who Loves AI Natural language:
“There exists at least one student who loves AI.”

$$\exists x (Student(x) \wedge Loves(x, AI))$$

Meaning: Some object x exists such that x is a student and x loves AI.

Example 4: Every Student Has Submitted an Assignment Natural language:
“Every student has submitted an assignment.”

$$\forall x (Student(x) \rightarrow Submitted(x, Assignment))$$

Meaning: For all objects x , if x is a student, then x has submitted an assignment.

Example 5: Some Cats are Black Natural language:
“There exists at least one cat that is black.”

$$\exists x (Cat(x) \wedge Black(x))$$

Meaning: There is at least one x such that x is a cat and x is black.

Example 6: Friends Relation Natural language:
“If Alice is a friend of Bob, then Bob is a friend of Alice.”

$$\forall x \forall y (Friend(x, y) \rightarrow Friend(y, x))$$

Meaning: Friendship is mutual between any two individuals.

Example (FOL):

$$\forall x (Cat(x) \rightarrow Mammal(x))$$

Meaning: All cats are mammals.

2. Semantic Networks

Semantic networks represent knowledge as a graph of nodes (concepts) connected by edges (relationships). They are intuitive and good for hierarchical relationships.

A **Semantic Network** is a knowledge representation technique that uses a graph of nodes and links to show how concepts are related. Nodes represent *objects*, *concepts*, or *events*, and links represent *relationships* between them.

Key Features

- Graph-based structure with nodes and arcs.
- Labels on nodes and edges indicate meaning.
- Often arranged hierarchically for inheritance of properties.

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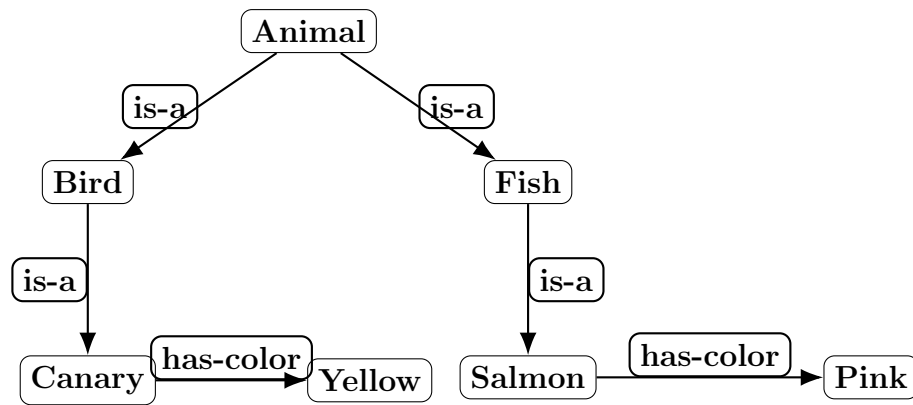


Figure 1: Example of a Semantic Network

Example of a Semantic Network

Example 1: Animal Hierarchy

- Canary is a Bird.
- Birds can fly and have wings.
- Canary is yellow and can sing.

In this network, the **Canary** node inherits properties from the **Bird** node, such as *can fly* and *has wings*.

Example 2: Family Relationship Network

- John is the father of Mary.
- Mary is the sister of Paul.
- Paul is the son of Sarah.

Advantages

- Clear visualization of relationships.
- Supports inheritance to reduce redundancy.
- Intuitive mapping to human memory models.

Limitations

- Can grow large and complex quickly.
- Lacks standardization for detail level.
- Not ideal for incomplete or uncertain data.

3. Frame Representation

Frames are data structures for representing stereotyped situations. They contain slots (attributes) and values that describe an object or concept. A frame is a data structure used in Artificial Intelligence to represent stereotyped situations or objects in terms of attributes (slots) and their corresponding values. Frames are an extension of the semantic network concept, but instead of just nodes and links, frames can store more detailed information, including default values, procedures, and constraints

Structure of a Frame

- **Frame Name** – the concept or object being represented.
- **Labels** on nodes and edges indicate meaning.
- **Slots** – attributes or properties of the object.

Example of a Frame

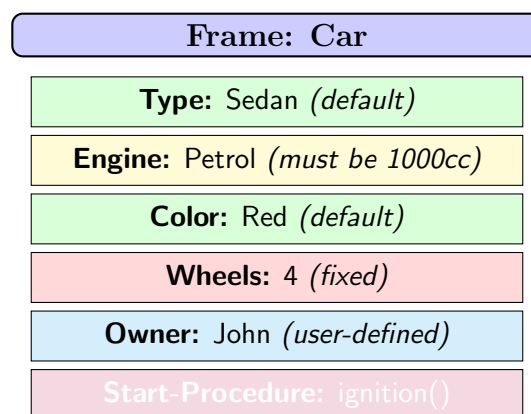


Figure 2: Styled Frame Representation for a Car

```
Frame: Cat
  Type: Animal
  Legs: 4
  Sound: "Meow"
  Color: Variable
```

Frames allow inheritance: If **Cat** is a subclass of **Mammal**, it inherits properties like “Warm-blooded”.

Advantages of Frame Representation

- **Structured Knowledge** – Provides a clear and organized representation.
- **Default Reasoning** – Uses default values when no specific information is given.

- **Inheritance** – Subframes can inherit properties from parent frames (similar to object-oriented programming).
- **Flexibility** – Can store facts, default assumptions, and procedures in one place.

4. Production Rules

Production rules are a way of representing knowledge in the form of **condition–action** pairs. They are widely used in expert systems and rule-based systems to express logic in the format:

IF condition THEN action

or

IF condition AND condition THEN action

They are sometimes called **IF-THEN rules** and are often used with an inference engine that decides which rule to apply based on the current state of knowledge.

Structure of a Production Rule

- **Condition (IF part)** – A statement or set of statements that must be true for the rule to fire.
- **Action (THEN part)** – The operation or conclusion that is applied if the condition is true.

Formally:

Rule: IF Premise(s) THEN Conclusion/Action

Example 1 – Medical Diagnosis

Rule 1: IF patient_has_fever AND patient_has_cough THEN diagnosis_is_flu.

Rule 2: IF patient_has_rash AND patient_has_fever THEN diagnosis_is_measles.

Example 2 – Game AI

Rule 1: IF enemy_is_near AND health_is_low THEN retreat.

Rule 2: IF enemy_is_near AND health_is_high THEN attack.

Working Mechanism

Production systems operate in a cycle:

1. **Match Phase** – Compare the current facts with the conditions of all rules to find applicable ones.
2. **Conflict Resolution** – If multiple rules can fire, choose one based on priority or specificity.
3. **Act Phase** – Apply the selected rule's action, which may add or remove facts.
4. Repeat until no more rules apply or a goal is reached.

Example 3 – Animal Identification

Facts:

Fact: has_feathers

Fact: can_fly

Rules:

R1: IF has_feathers THEN is_a_bird.

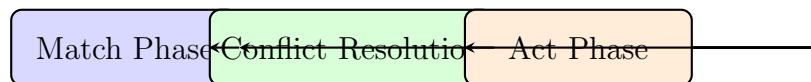
R2: IF is_a_bird AND can_fly THEN bird_type_is_flying.

R3: IF is_a_bird AND NOT can_fly THEN bird_type_is_flightless.

Reasoning process:

1. Match has_feathers → Fire R1 → Add is_a_bird.
2. Match is_a_bird AND can_fly → Fire R2 → Add bird_type_is_flying.
3. No further applicable rules → Stop.

Diagram of Production System Cycle



Compare facts with rules
Select one rule to fire
Execute action and update facts

Advantages

- Simple and modular.
- Easy to modify and extend.
- Suitable for domains where knowledge can be expressed as rules.

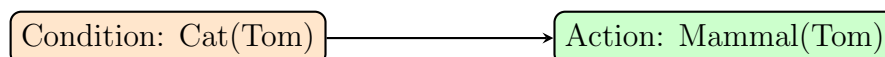
Disadvantages

- Can become inefficient with large rule sets.
- Conflict resolution can be complex.
- Difficult to represent uncertainty without extensions.

They are common in expert systems and allow step-by-step reasoning.

IF Cat(x) THEN Mammal(x)

IF Mammal(x) AND Domestic(x) THEN Pet(x)



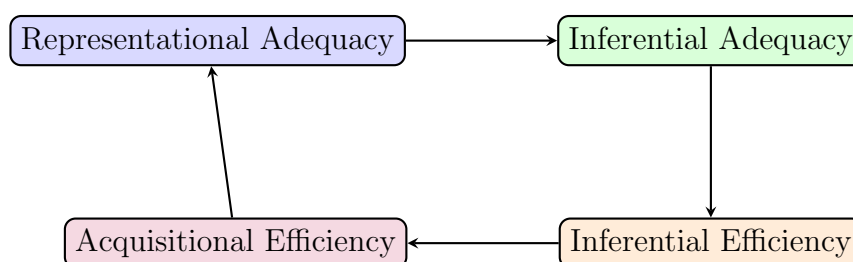
Comparison Table

KR Type	Advantages	Disadvantages	Example
Logical Representation	Precise, supports inference	Can be computationally expensive	Theorem proving
Semantic Networks	Easy to visualize	May lack formal semantics	Concept maps
Frames	Supports inheritance	Rigid structure	Object-oriented programming
Production Rules	Simple to implement	May grow large and unmanageable	Expert systems

Knowledge Representation Requirements

A good knowledge representation system should satisfy the following requirements:

1. **Representational Adequacy** It must be capable of representing all kinds of knowledge required in the domain.
2. **Inferential Adequacy** It should allow new knowledge to be inferred from existing facts.
3. **Inferential Efficiency** The representation should enable reasoning to be carried out efficiently, minimizing computational cost.
4. **Acquisitional Efficiency** It should be easy to acquire, update, and modify knowledge without major redesign.
5. **Clarity and Unambiguity** Knowledge should be expressed in a way that avoids misinterpretation.
6. **Support for Learning** The system should be able to incorporate new knowledge over time.

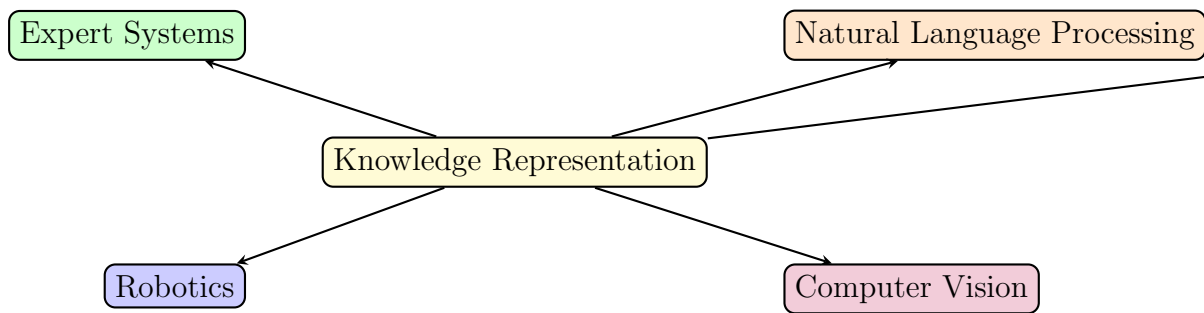


Applications of Knowledge Representation in AI

Knowledge representation is applied in many domains of artificial intelligence, including:

1. **Expert Systems** – Use production rules to diagnose problems and recommend solutions. *Example:* MYCIN (medical diagnosis system).
2. **Natural Language Processing (NLP)** – Represent linguistic knowledge for parsing, understanding, and generating human language. *Example:* Chatbots, translation systems.
3. **Robotics** – Store and reason about the physical world to plan movements and actions. *Example:* Path planning using semantic maps.

4. **Computer Vision** – Represent object features and relationships for recognition. *Example:* Scene understanding.
5. **Game AI** – Represent game states and possible moves for decision-making. *Example:* Chess engines.
6. **Semantic Web** – Encode data with meaning to enable automated agents to process web content. *Example:* RDF and OWL ontologies.



Summary

Knowledge Representation is the backbone of intelligent systems, determining how they perceive, reason, and act in the world. Choosing the right representation impacts both performance and scalability.